Experiment 6

Bad Data Detection and Elimination in MATLAB

Lab Report

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MATLAB code with comments

https://github.com/hamza-2274/PSOC/tree/main/badestimation

```
% Experiment 6: Bad Data Detection and Elimination
clear
clc
close all
%% ini data
V = [1.06; 1.045; 1.01; 1.01772371530645; 1.01956929175548; 1.07; ...
    1.06158474200345; 1.09; 1.05599579866432; 1.05104189972113; ...
```

```
    1.05694910281384; 1.05520754822189; 1.05040466613156;
1.03557823098035];

delta = zeros(14,1);

B = [0.0528 0.0492 0.0438 0.0340 0.0346 0.0128 0 0 0 0 0 0 0 0 0 0 0 0
0]';

X = [0.05917; 0.22304; 0.19797; 0.17632; 0.17388; 0.17103; 0.04211; 0.1989;
...

    0.25581; 0.13027; 0.17615; 0.11001; 0.0845; 0.27038; 0.19207; 0.19988;
...

    0.34802; 0.20912; 0.55618; 0.20912];

R = [0.01938; 0.05403; 0.04699; 0.05811; 0.05695; 0.06701; 0.01335;
0.09498; ...

    0.12291; 0.06615; 0; 0; 0.03181; 0.12711; 0.08205; 0.22092; 0.17093; 0;
0; 0];

from_bus = [1; 1; 2; 2; 2; 3; 4; 4; 4; 5; 6; 6; 6; 7; 7; 9; 9; 10; 12; 13];

to_bus = [2; 5; 3; 4; 5; 4; 5; 7; 9; 6; 11; 12; 13; 8; 9; 10; 14; 11; 13;
14];

Z = R + 1j*X;

y = 1 ./ Z;

Nbus = length(V);

branches = length(from_bus);

[Ybuss] = Ybuss();

[z_true] = SE_Measurments();

rng(42);

sigma = 0.02;

noise = sigma * randn(length(z_true), 1);

z = z_true + noise;

z_original = z;

%% inject bad data

bad_measurement_index = 50;

bad_data_magnitude = 0.15;

z(bad_measurement_index) = z(bad_measurement_index) + bad_data_magnitude;

fprintf('Bad data  measurement index: %d\n', bad_measurement_index);

fprintf('Bad data : %.4f\n', bad_data_magnitude);

fprintf('Original value: %.6f, Corrupted value: %.6f\n\n', ...

   z_original(bad_measurement_index), z(bad_measurement_index));
```

```
W = eye(length(z)) / (sigma^2);
%% chi-squared
alpha = 0.05;
m = length(z);
n = 2*Nbus - 1;
degrees_of_freedom = m - n;
chi2_threshold = chi2inv(1 - alpha, degrees_of_freedom);
fprintf('para:\n');
fprintf('  (alpha): %.2f\n', alpha);
fprintf('  m): %d\n', m);
fprintf('  n): %d\n', n);
fprintf('  Degrees of freedom: %d\n', degrees_of_freedom);
fprintf('   threshold: %.4f\n\n', chi2_threshold);
%% State Estimation with Bad Data Detection
iteration_outer = 0;
max_outer_iter = 10;
bad_data_detected = true;
bad_data_indices = [];
tau = 3.0;
tau_adaptive = 0.15;
active_measurements = true(length(z), 1);
while bad_data_detected && iteration_outer < max_outer_iter
   iteration_outer = iteration_outer + 1;


   fprintf('-----------------------------------------------------\n');
   fprintf('Outer Iteration %d: Running State Estimation\n',
iteration_outer);


   x = [delta(2:end); V];
   z_active = z(active_measurements);
   W_active = W(active_measurements, active_measurements);
   % WLS
   max_iter = 50;
```

```
    tolerance = 1e-4;
    iter = 0;
    converged = false;

    while iter < max_iter && ~converged
        iter = iter + 1;
        delta = [0; x(1:Nbus-1)];
        V = x(Nbus:end);
        [H_full] = State_Estimation_jacobian(V, delta, B, from_bus, to_bus,
Ybuss);
        [f_full] = SEFunc(V, delta, y, from_bus, to_bus, Ybuss);
        H = H_full(active_measurements, :);
        f = f_full(active_measurements);
        e = z_active - f;
        G = (H' * W_active * H) \ (H' * W_active * e);%corr
        x = x + G;
        max_mismatch = max(abs(G));

        if max_mismatch < tolerance
            converged = true;
        end
    end

    fprintf('  State estimation converged in %d iterations\n', iter);

    %% Compute Final Residuals
    delta = [0; x(1:Nbus-1)];
    V = x(Nbus:end);
    [H_full] = State_Estimation_jacobian(V, delta, B, from_bus, to_bus,
Ybuss);
    [f_full] = SEFunc(V, delta, y, from_bus, to_bus, Ybuss);

    % Sactive measurements
    H = H_full(active_measurements, :);
```

```
f = f_full(active_measurements);
r = z_active - f; % Residual vector

%% Chi-Squared Test
J_test = r' * W_active * r;

m_active = sum(active_measurements);
degrees_of_freedom = m_active - n;
chi2_threshold = chi2inv(1 - alpha, degrees_of_freedom);

fprintf('  Chi-squared test statistic J(x): %.4f\n', J_test);
fprintf('  Chi-squared threshold: %.4f\n', chi2_threshold);

if J_test > chi2_threshold
    fprintf('  Result: BAD DATA SUSPECTED (J > threshold)\n\n');

    % = H(H'WH)^(-1)H'
    Omega = H * ((H' * W_active * H) \ H');

   % S = I - ΩW
    S = eye(m_active) - Omega * W_active;

    r_normalized = zeros(m_active, 1);
    for i = 1:m_active
        if S(i,i) > 1e-10 % Avoid division by zero
            r_normalized(i) = abs(r(i)) / sqrt(S(i,i));
        else
            r_normalized(i) = 0;
        end
    end

    [max_r_norm, bad_index_local] = max(r_normalized);
```

```
        % Map
        active_indices = find(active_measurements);
        bad_index_original = active_indices(bad_index_local);

        %  top 5
        [sorted_r_norm, sorted_idx] = sort(r_normalized, 'descend');
        fprintf('  Top 5 suspicious measurements:\n');
        for k = 1:min(5, length(sorted_r_norm))
            meas_idx = active_indices(sorted_idx(k));
            fprintf('    Measurement %3d: r_N = %.4f, residual = %.6f\n',
...
                meas_idx, sorted_r_norm(k), r(sorted_idx(k)));
        end
        fprintf('\n');

        fprintf('  Largest normalized residual: %.4f at measurement %d\n',
max_r_norm, bad_index_original);

        % Adaptive stopping criterion based on Chi-squared test result
        J_ratio = J_test / chi2_threshold;

        fprintf('  Chi-squared ratio (J/threshold): %.2f\n', J_ratio);

        if max_r_norm > tau
            % Standard case: clear bad data with high normalized residual
            fprintf('  Action: REMOVING measurement %d (|r_N| = %.4f >
%.1f)\n\n', bad_index_original, max_r_norm, tau);
            active_measurements(bad_index_original) = false;
            bad_data_indices = [bad_data_indices; bad_index_original];

        elseif J_ratio > 1.2 && max_r_norm > tau_adaptive
            fprintf('  Action: REMOVING measurement %d (J/threshold = %.2f,
r_N = %.4f > %.2f)\n\n', ...
                bad_index_original, J_ratio, max_r_norm, tau_adaptive);
            active_measurements(bad_index_original) = false;
```

```
            bad_data_indices = [bad_data_indices; bad_index_original];


        else
            % No clear bad measurement
            if J_ratio > 1.1
                fprintf('  Warning: Chi-squared still elevated (%.2f × 
threshold) but no clear bad measurement\n', J_ratio);
                fprintf('  Action: Stopping to avoid removing good 
measurements\n\n');
            else
                fprintf('  Action: No bad measurement found. 
Stopping.\n\n');
            end
            bad_data_detected = false;
        end
    else
        fprintf('  Result: NO BAD DATA DETECTED (J <= threshold)\n\n');
        bad_data_detected = false;
    end
end
%% Final Results
fprintf('============================================================\n');
fprintf('   FINAL RESULTS (After Bad Data Removal)\n');
fprintf('============================================================\n\n');
fprintf('Bad measurements removed: %d\n', length(bad_data_indices));
if ~isempty(bad_data_indices)
    fprintf('Indices of removed measurements:\n');
    disp(bad_data_indices);
end
fprintf('\nFinal Estimated States:\n');
fprintf('Bus   Voltage (p.u.)   Angle (deg)\n');
fprintf('-----------------------------------------\n');
delta_deg = rad2deg([0; x(1:Nbus-1)]);
for i = 1:Nbus
```

```
    fprintf('%2d    %8.6f       %8.4f\n', i, V(i), delta_deg(i));
end
%% Run comparison with clean data (no bad data injection)
fprintf('\n=====================================================\n');
fprintf('   COMPARISON: Running with CLEAN data (no bad data)\n');
fprintf('=====================================================\n\n');
% Reset measurements to clean version (with noise but no bad data)
z_clean = z_original;
active_clean = true(length(z_clean), 1);
x_clean = [zeros(Nbus-1,1); [1.06; 1.045; 1.01; 1.01772371530645;
1.01956929175548; 1.07; 1.06158474200345; 1.09; 1.05599579866432;
1.05104189972113; 1.05694910281384; 1.05520754822189; 1.05040466613156;
1.03557823098035]];
% Run one state estimation
max_iter = 50;
tolerance = 1e-4;
iter = 0;
while iter < max_iter
    iter = iter + 1;

    delta_clean = [0; x_clean(1:Nbus-1)];
    V_clean = x_clean(Nbus:end);

    [H_clean] = State_Estimation_jacobian(V_clean, delta_clean, B, from_bus,
to_bus, Ybuss);
    [f_clean] = SEFunc(V_clean, delta_clean, y, from_bus, to_bus, Ybuss);

    e_clean = z_clean - f_clean;
    G_clean = (H_clean' * W * H_clean) \ (H_clean' * W * e_clean);
    x_clean = x_clean + G_clean;

    if max(abs(G_clean)) < tolerance
        break;
    end
end
```

```
% Compute Chi-squared for clean data

delta_clean = [0; x_clean(1:Nbus-1)];

V_clean = x_clean(Nbus:end);

[H_clean] = State_Estimation_jacobian(V_clean, delta_clean, B, from_bus,
to_bus, Ybuss);

[f_clean] = SEFunc(V_clean, delta_clean, y, from_bus, to_bus, Ybuss);

r_clean = z_clean - f_clean;

J_clean = r_clean' * W * r_clean;

fprintf('Clean data Chi-squared statistic: %.4f (threshold: %.4f)\n',
J_clean, chi2_threshold);

if J_clean <= chi2_threshold

   fprintf('Clean data result: PASS (no bad data detected)\n');

else

   fprintf('Clean data result: Still high (may need investigation)\n');

end

fprintf('Clean data converged in %d iterations\n\n', iter);

fprintf('Clean Data Estimated States:\n');

fprintf('Bus   Voltage (p.u.)   Angle (deg)\n');

fprintf('-----------------------------------\n');

delta_clean_deg = rad2deg([0; x_clean(1:Nbus-1)]);

for i = 1:Nbus

   fprintf('%2d    %8.6f       %8.4f\n', i, V_clean(i),
delta_clean_deg(i));

end

fprintf('\n=====================================================\n');

fprintf('   Algorithm completed successfully!\n');

fprintf('=====================================================\n');
```

RESULTS

Measurement set with at least one bad data injection.

Iteration table showing detection and elimination process.

Final estimated states and comparison with load flow.

1	0.886385
2	0.447025
3	0.412107
4	0.275047
5	0.306345
6	0.176644
7	0.176357
8	0.100400
9	0.081022
10	0.095399
11	0.059991
12	0.068664
13	0.040273
14	0.042106
15	0.022696
16	0.021048
17	0.019414
18	0.012686
19	0.015052
20	0.009004
21	0.009724
22	0.005387
23	0.005258
24	0.003830
25	0.002588
26	0.003192
27	0.001963
28	0.002193
29	0.001250

30	0.001269
31	0.000723
32	0.000589
33	0.000662
34	0.000419
35	0.000484
36	0.000283
37	0.000298
38	0.000161
39	0.000151
40	0.000133
41	0.000087

Command Window

```
    40           0.000133
    41           0.000087
  Bad data  measurement index: 50
  Bad data : 0.1500
  Original value: 0.216807, Corrupted value: 0.366807

  para:
    (alpha): 0.05
    m): 122
    n): 27
    Degrees of freedom: 95
     threshold: 118.7516

  ------------------------------------------------------
  Outer Iteration 1: Running State Estimation
    State estimation converged in 9 iterations
    Chi-squared test statistic J(x): 181.8598
    Chi-squared threshold: 118.7516
    Result: BAD DATA SUSPECTED (J > threshold)

    Top 5 suspicious measurements:
      Measurement  50: r_N = 0.1705, residual = 0.153180
      Measurement  70: r_N = 0.0586, residual = 0.052659
      Measurement  33: r_N = 0.0583, residual = 0.038939
      Measurement  23: r_N = 0.0539, residual = 0.035886
```

```
--------------------------------------------------
Outer Iteration 1: Running State Estimation
  State estimation converged in 9 iterations
  Chi-squared test statistic J(x): 181.8598
  Chi-squared threshold: 118.7516
  Result: BAD DATA SUSPECTED (J > threshold)

  Top 5 suspicious measurements:
    Measurement  50: r_N = 0.1705, residual = 0.153180
    Measurement  70: r_N = 0.0586, residual = 0.052659
    Measurement  33: r_N = 0.0583, residual = 0.038939
    Measurement  23: r_N = 0.0539, residual = 0.035886
    Measurement  77: r_N = 0.0521, residual = -0.045384

  Largest normalized residual: 0.1705 at measurement 50
  Chi-squared ratio (J/threshold): 1.53
  Action: REMOVING measurement 50 (J/threshold = 1.53, r_N = 0.1705 > 0.15)
```

```
--------------------------------------------------
Outer Iteration 2: Running State Estimation
  State estimation converged in 5 iterations
  Chi-squared test statistic J(x): 111.0153
  Chi-squared threshold: 117.6317
  Result: NO BAD DATA DETECTED (J <= threshold)
```

Command Window

```
   FINAL RESULTS (After Bad Data Removal)
================================================

Bad measurements removed: 1
Indices of removed measurements:
    50


Final Estimated States:
Bus    Voltage (p.u.)    Angle (deg)
----------------------------------------
 1     1.063151           0.0000
 2     1.048791          -4.9571
 3     1.012945         -12.5786
 4     1.025982         -10.2959
 5     1.034480          -8.8722
 6     1.075795         -14.5970
 7     1.058456         -13.4260
 8     1.097068         -13.4847
 9     1.041351         -15.0161
10     1.043198         -15.1541
11     1.056330         -14.9269
12     1.058241         -15.4491
13     1.052127         -15.5158
14     1.025301         -16.2687
```

```
Command Window
   COMPARISON: Running with CLEAN data (no bad data)
==================================================

Clean data Chi-squared statistic: 113.0679 (threshold: 117.6317)
Clean data result: PASS (no bad data detected)
Clean data converged in 9 iterations

Clean Data Estimated States:
Bus    Voltage (p.u.)    Angle (deg)
----------------------------------------
 1     1.063478          0.0000
 2     1.049117         -4.9545
 3     1.013324        -12.5681
 4     1.026190        -10.2993
 5     1.034712         -8.8736
 6     1.075760        -14.6199
 7     1.058411        -13.5033
 8     1.096902        -13.5874
 9     1.041323        -15.0808
10     1.043175        -15.2138
11     1.056286        -14.9674
12     1.058144        -15.4725
13     1.052054        -15.5423
14     1.025291        -16.3169
fx
```

Discussion of Performance and Limitations

Performance of the Algorithm

The implemented bad data detection and elimination scheme demonstrates **effective performance** under the tested conditions. By combining **Weighted Least Squares (WLS)** state estimation with a **chi-squared global test** and **largest normalized residual analysis**, the algorithm is able to:

1. **Correctly detect the presence of bad data**
   The injected gross error causes the chi-squared test statistic J(x)J(x)J(x) to exceed the theoretical threshold, indicating inconsistency between measurements and the estimated states. This confirms the sensitivity of the global test to abnormal measurement behavior.

2. **Accurately localize the bad measurement**
   The normalized residual method successfully identifies the corrupted measurement as having the largest residual magnitude. The use of the sensitivity matrix
   S=I−H(HTWH)−1HTWS = I - H(H^TWH)^{-1}H^TWS=I−H(HTWH)−1HTW

allows residuals to be properly scaled, improving discrimination between noisy but valid data and truly bad data.

3. **Robust convergence after bad data removal**
   Once the suspect measurement is removed, the state estimator reconverges within a few iterations, and the chi-squared test statistic falls below the threshold. This indicates that the remaining measurement set is statistically consistent.

4. **Consistency with clean data results**
   The comparison with the clean (non-corrupted) dataset shows similar voltage magnitudes and phase angles, confirming that the bad data elimination restores the estimation accuracy close to the nominal case.

5. **Controlled data rejection**
   The inclusion of an adaptive stopping criterion (based on both normalized residual magnitude and the chi-squared ratio) helps prevent unnecessary removal of valid measurements, improving algorithm stability.

Overall, the results validate the **reliability and effectiveness** of classical residual-based bad data processing when a single dominant bad measurement is present.

Limitations of the Approach

Despite its good performance, the implemented method has several important limitations:

1. **Sensitivity to multiple interacting bad data**
   The normalized residual method assumes that bad data affect measurements independently. When multiple bad measurements are present simultaneously—especially if they are correlated—the residuals may mask each other, leading to missed detection or incorrect removal.

2. **Dependence on accurate noise statistics**
   The chi-squared test relies on correct knowledge of the measurement noise variance (σ2\sigma^2σ2). If the assumed noise model is inaccurate, the test may produce false alarms or fail to detect bad data.

3. **Threshold selection is heuristic**
   The choice of residual thresholds (τ\tauτ, τadaptive\tau_{\text{adaptive}}τadaptive) is empirical. Different systems, noise levels, or redundancy ratios may require retuning, limiting the method's universality.

4. **Risk of removing good measurements**
   In low-redundancy systems, removing a valid measurement can significantly degrade observability or estimation accuracy. Although the adaptive stopping rule mitigates this risk, it cannot eliminate it entirely.

5. **Computational burden for large systems**
   The repeated WLS estimation and matrix inversions inside the outer bad data loop increase computational cost. While acceptable for this test system, scalability to large-scale networks may require optimized solvers or alternative formulations.

6. **Inability to detect leverage points**
   Measurements with high leverage (large influence on the state estimate) may produce small residuals even when corrupted. Classical residual-based methods are known to perform poorly in detecting such bad data.